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A Multi-Stream Deep Learning Framework for Driver State Monitoring

Kanika Dhawan¹, Prince Kushwaha¹, Sukhjot Singh Sehra^{2,*}, Jaiteg Singh³ and Sumeet Kaur Sehra⁴

¹ Sabudh Foundation, Mohali, India
² Wilfrid Laurier University, Waterloo, Canada
³ Chitkara University, Rajpura, India
⁴ Conestoga College, Waterloo, Canada
*Author to whom any correspondence should be addressed.

E-mail: ssehra@wlu.ca

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Abstract

Negligent driver behaviour caused by fatigue, distraction, and emotional stress has significantly increased in road accidents, which requires effective real-time monitoring solutions. This paper proposes a scalable, multi-modal driver state monitoring system that integrates three distinct assessments: driver attention, facial emotion recognition, and drowsiness detection, each handled by specialized deep learning architectures. Specifically, MobileNetV2, combined with physiological metrics such as the eye-aspect ratio, is used for drowsiness detection; Long Short-Term Memory (LSTM) networks with an attention mechanism evaluate driver attentiveness and ResNet18 is applied for facial expression recognition. Unlike traditional approaches that provide a single aggregated risk score, our approach produces three separate, interpretable scores that quantify attention, emotion, and drowsiness, thereby delivering a comprehensive and granular driver risk assessment. Experimental validation in benchmark datasets, including NTHU and FER2013, demonstrates robust performance, achieving an overall precision of approximately 90.8%. This multi-stream framework enables early detection of risky driving states with adaptability to diverse driving conditions, making it practical for real-world deployment on varied hardware platforms.

1 Introduction

Fatigue and inattentive driving are recognized as significant contributors to road crashes worldwide. These factors pose significant risks to public and traffic safety [1]. Drowsy driving contributes to between 3 and 30% of all traffic accidents worldwide [2], while distractions can significantly increase accident risk [3]. These statistics underscore the urgent need for effective real-time monitoring solutions.

Most driver monitoring systems focus on physiological features, such as heart rate measured by various sensors [4], or behavioral signals including eye blink rate, eyelid closure, head movements, and steering wheel patterns to assess driver states such as fatigue or distraction. Computer vision models frequently track eyelid closure, eye aspect ratio, yawning detection, and steering input anomalies. These systems typically employ classical machine learning models or neural networks to classify driver states. For example, Support Vector Machines (SVMs), Random Forests, and standard Convolutional Neural Networks (CNNs) are commonly used for facial landmark and blink detection. While these methods show effectiveness in controlled settings, their performance declines in real-world environments due to factors such as inconsistent lighting, facial occlusions (e.g., sunglasses), and inter-driver variability [5].

A significant limitation of these methods is their inability to capture the dynamic nature of driver behavior. Focusing exclusively on a single feature, such as eyelid closure or steering drift, may lead to missed signs of distraction, including abnormal gaze or head posture. While these sensor technologies are practical, they generally rely on a limited subset of facial or movement

features [6]. Recent research [7] has explored multi-modal approaches that combine visual, temporal, and physiological signals, utilizing deep learning architectures for more detailed analysis. However, these methods face challenges, including sensor cost, concerns about user privacy, computational overhead, and the reliability of real-time predictions in highly variable conditions.

This paper proposes a real-time driver monitoring system that simultaneously analyzes three key areas: driver attention, facial emotions, and drowsiness levels. The system integrates MobileNetV2, Long Short-Term Memory (LSTM) networks, and ResNet18 architectures to capture both spatial patterns and temporal changes from in-vehicle camera feeds. To improve reliability under real-world conditions, the system incorporates physiological features such as the eye aspect ratio (EAR). Testing on standard datasets, including NTHU and FER2013, demonstrates that the system achieves an accuracy of 90.8 %.

Unlike existing single-measurement systems, this unified approach independently analyzes multiple behavioral features, generating three distinct, interpretable outputs corresponding to attentiveness, emotion, and drowsiness. This triple-output design enables granular, targeted detection of risky driving states, enhancing interpretability and practical applicability. Furthermore, the system’s flexible architecture allows it to function on different hardware setups while accommodating diverse driving situations, thus enhancing its feasibility for real-world use.

The remainder of this paper is organized as follows: Section II elaborates on the existing systems. Section III provides the details of the datasets and pre-processing techniques employed; Section III outlines the architecture of the proposed models; Section IV presents the experimental findings; and Section V concludes with a summary and directions for future research.

2 Existing Systems

Accurately predicting driver behavior is essential for enhancing road safety and reducing accidents. Existing approaches leverage physiological, vehicular, and facial cues, but they often struggle to generalize across diverse drivers, vehicles, and driving conditions.

Driver behavior analysis typically involves physiological, vehicular, and facial features [8, 9]. Physiological signals such as heartbeat or heart rate variability are commonly recorded using sensors like ECG or wrist-worn devices to infer driver fatigue [10, 11]. However, these sensors often suffer from practical limitations including discomfort during prolonged use, slow response times, and reduced reliability under real-world driving conditions. To address these drawbacks, various machine learning and computer vision techniques have been explored to analyze driver behavior more effectively [12].

Previous research has investigated vehicle-embedded strategies incorporating a range of external driving parameters, such as vehicle speed, steering angle, acceleration, driver sleep efficiency, and traffic conditions, alongside visual features such as eye tracking [13]. These diverse inputs have been utilized within multiple machine learning frameworks including Logistic Regression, Random Forest, SVM, Naive Bayes, and Gated Recurrent Units (GRU). Among these, GRU models have demonstrated superior accuracy, reaching between 81 percent and 84 percent. Furthermore, hybrid approaches combining vehicular telemetry and physiological signals with deep learning models like CNN and Bidirectional Long Short-Term Memory networks (BiLSTM) have been employed for fatigue detection and aggressive driving prediction [14].

Behavior-based methodologies, while leveraging physiological and vehicular parameters, often lack sufficient generalization across heterogeneous driver populations, vehicle models, and driving conditions [15]. In particular, these techniques can fail to capture individual driver states such as fatigue or distraction effectively, which are critical for a holistic understanding of driver behavior [16]. To overcome these issues, facial cue-based detection methods have been developed, utilizing indicators such as yawning, eye closure, and head nodding through classical image processing combined with machine learning classifiers such as SVM or Artificial Neural Networks (ANN). For example, facial expression tracking integrated with temporal machine learning classifiers has shown promise in predicting unsafe driving behavior by identifying pre-incident facial feature patterns [17]. However, approaches relying on the Viola-Jones face detection algorithm and basic neural networks face challenges under low illumination or when drivers wear glasses or masks, leading to decreased accuracy. Moreover, many traditional methods have limited capacity for processing dynamic temporal changes in facial expressions, which hinders real-time applicability due to computational constraints.

The advancement of facial landmark detection models, including Dlib’s 68-point model, has facilitated the computation of eye aspect ratio metrics to quantitatively assess eye closure and blink rates as drowsiness indicators. FengYou(2019) [18] employed an SVM-based alertness classification using eye landmarks but reported significant performance degradation under poor lighting and

partial occlusions, caused by head movements or eye wear.

To further improve detection accuracy and efficiency, hybrid models have been proposed [19]. For instance, a fusion of facial landmarks and a modified YOLOv5 architecture achieved over 95 percent accuracy on benchmark datasets, demonstrating robustness to variations in head pose, illumination, and eye wear occlusion. The integration of YOLO-based detectors and facial landmark extraction has substantially enhanced real-time drowsiness and fatigue detection capabilities.

Despite these advances, behavior-based techniques commonly encounter two major challenges [20], including, high computational complexity and difficulty in effectively modeling dynamic temporal changes in facial expressions and fatigue levels.

Recent years have witnessed significant progress in employing multi-modal and transformer-based deep learning architectures for driver monitoring. These methods leverage multiple data sources such as face video, audio, and environmental context to enhance robustness and accuracy [21, 22]. Transformer networks, with their superior ability to model long-range spatial-temporal dependencies, have shown promise in capturing complex driver behaviors beyond what traditional CNNs and RNNs can achieve [8].

Multi modal fusion methods combine features from heterogeneous sensors, including cameras, microphones, and vehicle telematics, enabling richer behavioral understanding and more reliable real-time assessment [23, 24]. These models facilitate concurrent prediction of multiple driver states such as attention, drowsiness, and emotion within unified frameworks.

However, despite their superior performance, transformer based multi-modal systems often behave like "black boxes," providing risk scores without straightforward interpretability [22]. Such complexity can hinder understanding and trust in critical safety applications. In contrast, our approach emphasizes interpretability by generating a structured triplet of results that correspond directly to distinct behavioral states —attention, drowsiness, and emotion —while maintaining computational efficiency suitable for embedded deployment.

3 Proposed Methodology

This paper proposes a comprehensive system leveraging deep learning and computer vision techniques. Unlike isolated models, our approach integrates multiple behavioral features—driver drowsiness, visual attention, and emotional state—into a unified framework. This integration enables continuous real-time assessment of driver risk with increased precision and responsiveness.

At the core of our methodology is a set of specialized models, each tailored to analyze specific aspects of driver behavior. These models employ advanced neural network architectures and are trained and rigorously validated using benchmark data sets such as FER2013 and NTHU. By combining spatial features extracted from individual video frames with temporal patterns learned from sequential data, our system effectively captures dynamic driver states, including slow eye closure, inconsistent gaze, and fleeting facial expressions.

This integrated solution not only enhances the precision of detecting unsafe driver conditions but also offers a scalable and modular platform suitable for deployment in modern vehicles and intelligent transportation systems.

3.1 Data Description

For building a reliable driver monitoring system, the choice of datasets plays a central role. Since driver behavior includes multiple states such as drowsiness, attention, and emotion, diverse benchmark datasets are required to capture these variations under different conditions, including varying lighting environments and driver demographics, to ensure the models generalize well. For this reason, our system uses several well-known benchmark datasets, including Facial Expression Recognition (FER) 2013 for facial emotion recognition, and NTHU and Yawning Detection Dataset (YawDD) for driver drowsiness and attention analysis.

3.1.1 Driver Facial Expression Datasets: The FER2013 dataset [25] is extensively utilized for training and evaluating models in facial emotion recognition. It comprises 35,887 grayscale facial images, each resized to 48×48 pixels, categorized into seven emotion classes: angry, disgust, fear, happy, sad, surprise, and neutral. The dataset is partitioned into three subsets, with 28,709 images allocated for training, and 3,589 images each for public and private testing. Notably, the dataset exhibits significant class imbalance, where the "Disgust" category is the least frequent and "Happy" is the most prevalent.

To mitigate bias caused by this imbalance, stratified sampling was employed to create training and validation splits in an 80:20 ratio, preserving the proportional distribution of each emotion class across both subsets. The images are centrally aligned and normalized to ensure consistency



Figure 1. Sample frames from the FER-2013 Dataset

during analysis. Despite certain inherent limitations, FER2013 is one of the most widely adopted benchmarks in machine learning and computer vision research for emotion classification tasks.

3.1.2 Driver Drowsiness Datasets: YawDD [26], developed by the Distributed Collaborative Virtual Environments Research Laboratory (DISCOVER Lab) at the University of Ottawa, comprises two distinct subsets: ‘Dash’ and ‘Mirror’. The Dash dataset includes videos from 29 participants (13 females and 16 males), recorded with a dashboard-mounted camera under varying illumination conditions. Each participant is captured during normal driving, talking, and yawning states, contributing to a total of 22 annotated videos representing diverse facial characteristics.

The Mirror dataset consists of 322 videos from 110 participants, recorded via a front mirror-mounted camera. Each participant contributes three to four videos showcasing different conditions such as normal driving, talking or singing, and yawning. All videos were recorded at a resolution of 640×480 pixels in 24-bit RGB color at 30 frames per second and stored in AVI format without audio. For subsequent facial landmark detection and temporal analysis, frames were converted to RGB as needed and resized to standardized input dimensions during pre-processing.

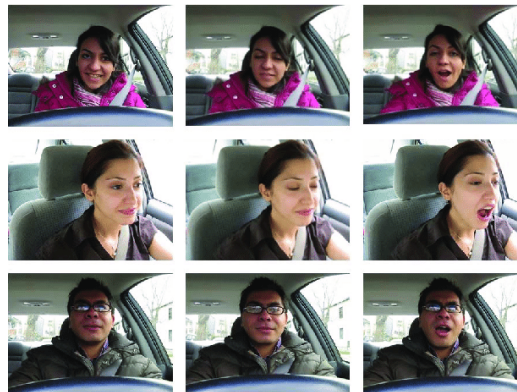


Figure 2. Sample frames from the Yawdd Dataset

The NTHU Drowsiness Dataset (NTHU-DDD) [27] is a well-established academic resource compiled by the Computer Vision Laboratory at National Tsing Hua University, Taiwan. It consists of approximately nine and a half hours of video footage recorded by a high-resolution camera under active infrared illumination, producing grayscale videos at 640×480 resolution in AVI format. The dataset includes recordings of 36 subjects from diverse ethnic backgrounds, each captured twice—once wearing glasses or sunglasses and once without—across varied simulated driving conditions.

These scenarios include normal driving, slow blinking, yawning, falling asleep, and bursting into laughter, simulating challenging real-world driving behaviors during both daytime and nighttime. Recordings were made under five conditions: Bare Face, Glasses, Sunglasses, Night-Bare Face, and Night-Glasses. The first three scenarios were recorded at 30 frames per second (fps), while the nighttime scenarios were captured at 15 fps. Participants simulated driving behaviors while seated in car seats and following specific facial expression instructions.

Subsequent extraction of facial features from these video frames has facilitated the development and evaluation of robust driver drowsiness detection models. NTHU [27] dataset is used as Driver Attention Dataset.



Figure 3. Sample frames from the NTHU Dataset

3.2 Data Pre-processing

Data pre-processing is a critical step in preparing raw datasets for the effective training of deep learning models in driver monitoring systems. This step involves cleaning and organizing the data to ensure consistency, structure, and suitability for efficient model learning. Although pre-processing steps vary depending on dataset characteristics, common tasks include resizing images or video frames to a uniform resolution, converting color channels to the format required by the network, and normalizing pixel intensity values to maintain a consistent numerical scale.

To extract meaningful cues related to driver behavior, facial landmark detection techniques are employed to analyze spatial features such as eye movements and head pose variations. Since driver behavior evolves over time, video frames are organized into temporal sequences to capture these dynamic changes. Additionally, when dividing data into training and testing sets, stratified sampling is utilized to preserve class distributions, thereby reducing bias in the model.

By carefully implementing these pre-processing steps, we ensure the input data is enriched with relevant spatial and temporal information, which is essential for accurately analyzing various aspects of driver behavior.

3.2.1 Dataset Pre-processing for Drowsiness Model: For developing a robust driver drowsiness detection model, careful pre-processing of input datasets is essential. This study employs two prominent datasets, YawDD and NTHU, each presenting distinct data characteristics and labeling approaches. To maintain consistency and leverage their complementary strengths, a unified pre-processing pipeline was applied to both.

The YawDD dataset consists of videos categorized by camera type (Mirror and Dash) and driver gender. Each video is processed by extracting frames sequentially at the original frame rate. Frames are resized to 320×240 pixels and converted from grayscale or BGR to RGB to standardize the input across all samples.

Facial landmark detection is performed on each frame using the MediaPipe Face Mesh, which outputs 468 normalized landmark coordinates. These landmarks enable the precise localization of facial regions. Specifically, a bounding box is computed around the landmarks and expanded by 20 pixels to ensure complete facial coverage. The bounding box coordinates are given by:

$$\begin{aligned} x_{\min} &= \max(0, \min_j(x_j) \cdot W - p), \\ x_{\max} &= \min(W, \max_j(x_j) \cdot W + p), \\ y_{\min} &= \max(0, \min_j(y_j) \cdot H - p), \\ y_{\max} &= \min(H, \max_j(y_j) \cdot H + p), \end{aligned}$$

where W and H denote the width and height of the frame, respectively, and $p = 20$ pixels is the padding.

Labeling frames in the YawDD dataset depends on camera type. For Mirror videos, yawning labels are extracted directly from the video filenames, where frames are labeled as yawning (1) or alert (0). For Dash videos, mouth openness is measured as the vertical distance between landmarks 13 and 14:

$$D = |y_{13} - y_{14}|,$$

with frames labeled as yawning if $D > 0.1$ and alert otherwise.

The cropped facial regions are resized to 224×224 pixels to comply with the input size requirements of the deep learning model. To capture temporal dynamics associated with drowsiness, frames are grouped into non-overlapping sequences of length 10:

$$S_k = \{F_k, F_{k+1}, \dots, F_{k+9}\}.$$

Pixel intensity values are normalized to the range $[0, 1]$ by dividing by 255, facilitating stable training.

The NTHU dataset similarly provides videos recorded under various simulated driving conditions. Frames are also resized to 320×240 pixels and converted to RGB. MediaPipe Face Mesh detects landmarks allowing face cropping with 20 pixels of padding. Unlike YawDD, frame-level drowsiness labels are supplied by the dataset, removing the need for manual labeling.

Frames from NTHU are organized into non-overlapping sequences of 10 frames, normalized identically to YawDD, and saved as labeled `.npz` files to support model fine-tuning.

Algorithm 1 summarizes the key pre-processing steps, ensuring structured data preparation that balances spatial and temporal information critical for accurate driver drowsiness detection.

Algorithm 1 Pre-processing Pipeline for Drowsiness Model

```

1: for each dataset in {YawDD, NTHU} do
2:   for each video file in dataset do
3:     Extract frames sequentially
4:     Resize frames to (320, 240) and convert to RGB
5:     Perform facial landmark detection (MediaPipe Face Mesh)
6:     Compute bounding box around landmarks with 20-pixel padding
7:     Crop facial region and resize to (224, 224)
8:     if dataset = YawDD then
9:       if camera type = Mirror then
10:        Assign label based on filename (1 = yawning, 0 = alert)
11:       else
12:        Compute mouth openness  $D = |y_{13} - y_{14}|$ 
13:        Label frame as 1 if  $D > 0.1$ , else 0
14:       end if
15:     else if dataset = NTHU then
16:       Use provided frame labels
17:     end if
18:     Group frames into sequences of length 10
19:     Normalize pixel values by dividing by 255
20:     Save each sequence as a labeled .npz file
21:   end for
22: end for

```

3.2.2 Dataset Processing for Driver Attention Model: The NTHU dataset is pre-processed to extract meaningful temporal features relevant to driver attention analysis. For each video frame, facial landmarks are detected using the MediaPipe Face Mesh framework. Specifically, five key points are identified per frame: the nose tip, the centers of the left and right eyes, and the centers of the left and right irises. The eye center and iris center coordinates are then calculated by averaging the corresponding left and right landmarks.

Gaze and head direction vectors are derived as the normalized differences between the iris and eye centers, and between the nose tip and eye centers. The resulting four-dimensional vector $[g_x, g_y, h_x, h_y]$ encapsulates both spatial and directional information, providing a compact yet informative representation of the driver's focus for each frame.

These frame-wise feature vectors are grouped into non-overlapping sequences of 30 frames to capture the short-term temporal dynamics of driver behavior. Each sequence is labeled as either *focused* or *distracted*, consistent with the driver's documented state. The sequences and associated labels are saved for model training, enabling the network to learn from both spatial configurations and temporal variations in attention.

Algorithm 2 presents a pre-processing strategy that effectively condenses raw video data into structured, high-level features, facilitating accurate and efficient driver attention classification within the proposed monitoring system.

3.2.3 Data Pre-processing for Facial Expression Model The pre-processing pipeline for the facial expression recognition model focuses on extracting high-quality, standardized facial images suitable

Algorithm 2 pre-processing Pipeline for Driver Attention Model

Require: Video frames $F = \{f_1, f_2, \dots, f_T\}$, sequence length $L = 30$, labels $y \in \{0 \text{ (Distracted)}, 1 \text{ (Focused)}\}$

Ensure: Sequences of feature vectors labeled by driver attention state

```

1: Initialize empty feature sequence
2: for each frame  $f_t \in F$  do
3:   Detect facial landmarks using MediaPipe FaceMesh
4:   Extract key points: nose  $P_n$ , left eye  $P_{le}$ , right eye  $P_{re}$ , left iris  $P_{li}$ , right iris  $P_{ri}$ 
5:   Compute eye center  $C_{eye} = \frac{P_{le} + P_{re}}{2}$  and iris center  $C_{iris} = \frac{P_{li} + P_{ri}}{2}$ 
6:   Calculate normalized gaze vector  $\mathbf{g} = \frac{C_{iris} - C_{eye}}{\|C_{iris} - C_{eye}\| + \epsilon}$ 
7:   Calculate normalized head vector  $\mathbf{h} = \frac{P_n - C_{eye}}{\|P_n - C_{eye}\| + \epsilon}$ 
8:   Form feature vector  $\mathbf{x}_t = [g_x, g_y, h_x, h_y]$ 
9:   Append  $\mathbf{x}_t$  to the current feature sequence
10:  if length of feature sequence ==  $L$  then
11:    Store the sequence  $X = \{\mathbf{x}_{t-L+1}, \dots, \mathbf{x}_t\}$  with label  $y$ 
12:    Reset the feature sequence
13:  end if
14: end for
15: Save all sequences and corresponding labels for training purposes

```

for deep learning. Raw images and video frames are processed with the MediaPipe Face Detection framework, which robustly localizes facial regions across varied lighting and pose conditions.

Detected faces are cropped from the original frames and resized to 48×48 pixels to balance computational efficiency and feature richness. To enhance model convergence and generalization, images are further converted to grayscale, thereby reducing color variability without compromising essential texture and shape information. Intensity normalization is performed by scaling pixel values to the $[-1, 1]$ range using mean and standard deviation of 0.5, standardizing inputs for stable gradient updates during training.

Data augmentation techniques, such as random horizontal flips and small rotations, are applied to artificially increase dataset diversity, mitigating overfitting risks given the limited size of facial emotion datasets. Labeled images are organized into standard emotion categories, including *Angry*, *Disgust*, *Fear*, *Happy*, *Neutral*, *Sad*, and *Surprise*, reflecting common affective states relevant in driver monitoring scenarios. This carefully designed pre-processing sequence ensures that the model receives consistent, normalized, and augmented facial data, improving the robustness and accuracy of emotion classification tasks.

Algorithm 3 pre-processing Pipeline for Facial Expression Model

Require: Raw video frames or images

Ensure: pre-processed, normalized, and augmented facial image dataset for training

```

1: for each frame/image  $f_t$  do
2:   Detect face regions using MediaPipe Face Detection
3:   if face detected then
4:     Crop face region from frame  $f_t$ 
5:     Resize cropped face to  $48 \times 48$  pixels
6:     Convert image to grayscale
7:     Normalize pixel intensity values to range  $[-1, 1]$ 
8:     Apply data augmentation (random horizontal flip, small rotations)
9:     Assign emotion label based on dataset annotation
10:    Store pre-processed and labeled image for training
11:  end if
12: end for

```

3.3 Model Architecture

The proposed driver monitoring system uses a hybrid deep learning architecture comprising multiple models, each specialized in detecting a specific aspect of driver behavior. This modular framework integrates Convolutional neural networks (CNNs) such as MobileNetV2 and ResNet18 to extract spatial features from facial images, alongside recurrent neural networks (RNNs) such as

Long Short-Term Memory Units (LSTM) to capture relevant temporal dependencies. Using transfer learning and selective layer fine-tuning, the architecture balances effective feature representation and computational efficiency. Each sub-model independently produces its own output of behavioral state: drowsiness, attentiveness, or emotion, resulting in a structured triplet of driver state indicators for every time window rather than a single fused safety score. This design supports real-time monitoring and flexible deployment in diverse driving scenarios. The following subsections detail the architecture and operation of each sub-model, as well as how they jointly contribute to the overall insight provided by the system.

3.3.1 Driver Drowsiness Architecture: The proposed driver drowsiness detection model employs a hybrid deep learning architecture that effectively fuses spatial and temporal information to capture behavioral patterns indicative of fatigue. MobileNetV2, a lightweight convolutional neural network pretrained on ImageNet, serves as the backbone for spatial feature extraction, processing individual video frames resized to 224×224 pixels after removing its original classification layers. To leverage transfer learning and prevent early overfitting, MobileNetV2 layers are initially frozen, allowing the model to retain learned low-level features while focusing on training the temporal components.

Temporal dependencies across sequences of 10 consecutive frames are modeled using a Long Short-Term Memory (LSTM) network with 64 units, equipped with dropout regularization to reduce the risk of overfitting. The spatial features from MobileNetV2 are first condensed by a Global Average Pooling 2D layer, producing compact frame-level embeddings that are fed sequentially into the LSTM to learn time-dependent patterns of drowsiness. The LSTM output passes through a fully connected dense layer with sigmoid activation, producing a probability score for binary classification of driver state (drowsy or alert). The model is trained using the Adam optimizer and binary cross-entropy loss.

Initial training is performed on the YawDD dataset, enabling the model to learn generalizable features associated with drowsiness, followed by fine-tuning on the more diverse NTHU dataset to adapt to varied lighting and driving conditions, thereby improving accuracy. During fine-tuning, the last several layers of MobileNetV2, along with the LSTM and dense layers, are unfrozen to enable weight updates. Training is performed at a reduced learning rate with callbacks such as early stopping and model check pointing to ensure convergence and prevent overfitting.

This architecture balances effective feature representation and computational efficiency, enabling real-time processing suitable for practical deployment. While the incorporation of MobileNetV2 and LSTM follows established approaches in driver monitoring, the novelty of this work lies in its integration within a unified multi-modal system that evaluates drowsiness, attentiveness, and emotional state, culminating in a comprehensive driver safety score. The design emphasizes transfer learning across datasets and employs carefully tuned temporal windows and voting schemes to enhance robustness. Future work will focus on generalization to unseen data, ablation studies, and incorporating additional multi-modal inputs for resilience under challenging conditions.

Experiments were conducted on GPU-enabled hardware using TensorFlow, with hyperparameter such as batch size, learning rate schedules, and sequence lengths reported to facilitate reproduction. The model architecture is illustrated in Figure 4.

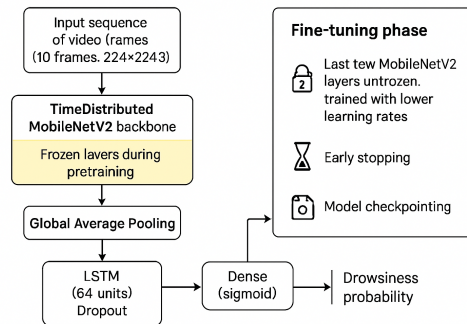


Figure 4. Algorithmic flowchart of the drowsiness model architecture

3.3.2 Driver Attention Model Architecture: The proposed driver attention model utilizes temporal sequences of head pose and gaze direction features to determine the driver's level of focus.

Raw video data are first processed to compute key facial landmarks using MediaPipe FaceMesh, with particular emphasis on eye, iris, and nose positions. For each frame, gaze and head direction vectors are generated and normalized, efficiently summarizing the driver's orientation and visual attention. These spatial descriptors are concatenated into four-dimensional feature vectors and stacked into input sequences of thirty frames, enabling the model to capture evolving behavioral cues.

The feature sequence is then processed by a multi-layer Long Short-Term Memory (LSTM) network, which excels at modeling the intricate temporal dependencies characteristic of driver attention patterns. To further enhance interpretability and focus on the most informative segments, an attention mechanism is incorporated, dynamically weighting the contribution of each frame's hidden representation in forming the context vector for classification. This context vector is then passed through a fully connected layer and softmax activation to produce probability scores for focused and distracted states. Training occurs with cross-entropy loss and an Adam optimizer, and hyperparameter such as the sequence length and hidden dimensions are empirically optimized for robust performance.

As illustrated in Figure 5, this pipeline allows the system to adaptively emphasize critical moments in a driving sequence, yielding accurate and interpretable attention assessment in real-world scenarios.

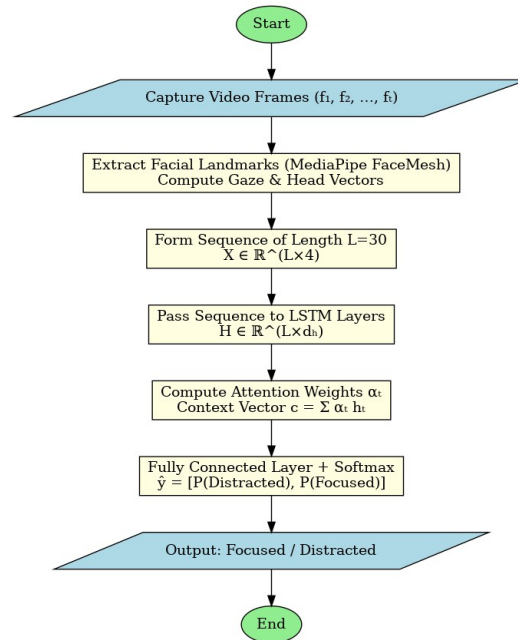


Figure 5. Algorithmic flowchart of the driver attention classification model.

3.3.3 Facial Expression Model Architecture: The facial expression recognition model is built upon the ResNet18 architecture, leveraging transfer learning from its pre-training on the large-scale ImageNet dataset. This backbone network provides a powerful hierarchy of convolutional layers that extract rich feature representations from facial images. To adapt ResNet18 for emotion classification, the original fully connected layer designed for 1000 ImageNet classes is replaced with a linear layer mapping the extracted feature vector $h \in \mathbb{R}^d$ to logits $z \in \mathbb{R}^C$, where $C = 7$ corresponds to the seven emotion classes considered in this work (Angry, Disgust, Fear, Happy, Sad, Surprise, and Neutral). Specifically, the logits are computed as

$$z = W_f h + b_f,$$

with $W_f \in \mathbb{R}^{C \times d}$ as the classification weight matrix and $b_f \in \mathbb{R}^C$ as the bias vector.

These logits are converted into class probabilities using the softmax function, allowing probabilistic interpretation of the model's predictions. Training is guided by minimizing the cross-entropy loss over batches, effectively penalizing divergence from the ground truth labels encoded in a one-hot format. To handle class imbalance present in facial emotion datasets, a weighted cross-entropy loss is optionally employed, assigning higher weights to underrepresented classes, thus encouraging the model to learn more balanced feature representations.

The model parameters are optimized using the Adam optimizer, with a learning rate schedule that reduces the rate when validation accuracy plateaus, thereby promoting stable and efficient convergence. Early stopping is used during training to prevent overfitting, halting the process when no improvement is observed within a predetermined patience window.

Initially, the ResNet18 backbone layers are frozen, focusing training on the new fully connected classification layer to leverage the pretrained features. Subsequent fine-tuning unfreezes selected convolutional layers, enabling adaptation to domain-specific nuances in driver facial expressions at a lower learning rate. This staged approach, illustrated in Figure 6, balances leveraging existing knowledge and adapting to task-specific characteristics, improving the model's generalization and accuracy in facial emotion recognition within driver monitoring scenarios.

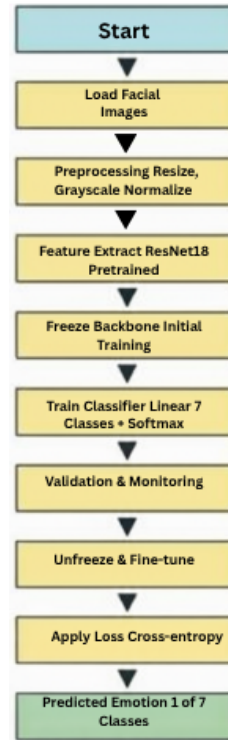


Figure 6. Algorithmic flowchart of the Driver Facial Expression Recognition model.

3.3.4 Driver State Multi-Output Representation: The unified driver monitoring framework presented in this study distinguishes itself from previous multi-modal approaches by integrating three advanced deep learning modules: attentiveness (using FocusLSTM with attention), drowsiness (via CNN-LSTM or eye aspect ratio thresholds), and facial emotion recognition (based on ResNet architectures) into a cohesive, parallel pipeline. This integration produces a comprehensive and interpretable representation of the driver's state. Unlike traditional multi-stream systems that analyze behavioral cues independently or rely on heuristic risk combinations, our approach synchronizes predictions within temporal windows, typically every 30 frames, corresponding to about one second of driving.

Within each temporal window, features such as gaze and head pose vectors, sequences of eye aspect ratio, and emotion-labeled facial crops are extracted and processed by their respective models. The attentiveness module employs a temporal attention mechanism to emphasize the most relevant frames, the drowsiness component combines handcrafted features with deep temporal modeling, and the facial emotion model applies majority voting across frame-level predictions to robustly determine the predominant affective state.

Instead of reducing these outputs to a single scalar safety score, our fusion method constructs a structured triplet:

$$DriverState = \{A, D, E\}$$

where A represents attentiveness as Focused or Distracted, D indicates drowsiness as Alert or Drowsy, and E categorizes emotion into one of Neutral, Happy, Sad, Surprise, Angry, Disgust, or

Fear. This design promotes enhanced interpretability and granular monitoring, offering a transparent alternative to existing multi-modal systems that often employ opaque fusion techniques.

The aggregation procedure is as follows: for every temporal window, outputs from the three models are consolidated, applying temporal voting or thresholding where applicable (especially for emotion and drowsiness classifications). The resulting triplet informs visualization, statistical summaries, and alert triggers. Pseudocode describing the fusion process is provided in Algorithm 4, illustrating the method's transparency and reproducibility. The overall architecture from raw video input to model pre-processing, to fused DriverState output is depicted in Figure 7.

Algorithm 4 Driver State Multi-Output Representation

```

1: for each temporal window do
2:    $A \leftarrow \text{FocusLSTM.predict}(\text{window})$ 
3:    $D \leftarrow \text{DrowsinessModel.predict}(\text{window})$ 
4:    $E \leftarrow \text{MajorityVote}(\text{EmotionModel.predict}(\text{frames in window}))$ 
5:    $\text{DriverState} \leftarrow \{A, D, E\}$ 
6:   Store  $\text{DriverState}$  for visualization and alerts
7: end for
  
```

Special attention has been given to domain adaptation and rigorous evaluation for experimental reproducibility. Models were trained and validated on distinct subject sets, with cross-validation and stratified sampling techniques employed to reduce bias. Detailed hyperparameter settings including learning rates, batch sizes, optimizers, training epochs, early stopping criteria, and class imbalance handling are comprehensively documented. Benchmarking was performed on NVIDIA RTX-series GPUs, with inference implemented using TensorFlow and PyTorch, enabling real-time processing typically exceeding 30 frames per second on standard hardware.

To support open science, code and pre-trained models will be publicly released along with scripts covering all pre-processing, training, and evaluation procedures. By emphasizing transparent fusion techniques, robust benchmarking, and reproducible practices, this work contributes to advancing the state of the art in multi-modal driver monitoring systems within Intelligent Transportation Systems.

Recent developments in driver monitoring have investigated transformer architectures and multi-modal fusion methods capable of capturing intricate temporal and cross-modal interactions. While these transformers excel at modeling complex dependencies, they often require substantial computational resources and tend to behave as black boxes, limiting interpretability an essential factor in safety-critical contexts like driver monitoring. Our framework, combines lightweight convolutional backbones with temporal LSTM and attention mechanisms within a modular design optimized for efficient real-time deployment. Our fusion strategy, based on synchronized temporal voting and structured triplets, delivers transparency and flexibility absent in conventional transformer-based approaches. This balance of performance, efficiency, modularity, and explainability positions our system favorably for practical applications in intelligent transportation.

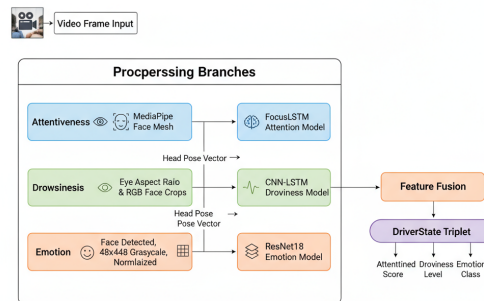


Figure 7. Unified driver monitoring pipeline illustrating the flow from video input through task-specific pre-processing, individual behavioral models, and fusion into a structured driver state output.

3.4 Ablation Study Plans

Although the current study demonstrates strong performance for the proposed multi-output framework, explicit ablation studies isolating the impact of key design choices are planned for

future work. These studies are critical to quantitatively assess the contribution of each module and the fusion strategy. Systematic experiments will evaluate:

- Removing or replacing temporal attention layers in driver attention modeling.
- Substituting backbone feature extractors (e.g., simpler CNNs) to assess trade-offs in accuracy, F1-score, and computational efficiency.
- Comparing the proposed voting-based aggregation to alternative fusion mechanisms and directly evaluating the effect of reporting separate outputs versus a fused score.
- Varying temporal sequence lengths to optimize behavioral context, tracking performance as a function of window size.
- Removing one behavioral sub-model at a time (“leave-one-out”) to measure the contribution of each output (attention, drowsiness, emotion).
- Comparing each sub-model’s performance to basic baseline models (e.g., SVM, Random Forest, or simple CNN).

For every ablation setting, metrics such as overall accuracy, F1-score, per-class recall, and inference speed will be reported, including mean and standard deviation across multiple random seeds or cross-validation folds to establish statistical significance. These analyses will help validate design decisions, highlight the unique utility of each behavioral sub-model, and guide iterative improvements. These analyses will help validate design decisions, highlight the unique utility of each behavioral sub-model, and guide iterative improvements for a robust and efficient real-time driver monitoring system.

4 Results

Building upon the modular framework introduced in the methodology, the proposed driver monitoring system employs three independently trained specialized models, each optimized on benchmark datasets specific to its task domain. The drowsiness detection model was trained on the YawDD and NTHU datasets; the attention classification model used the NTHU dataset; and the facial emotion recognition model was trained on the FER2013 dataset. Each of these models demonstrated consistently high accuracy, underscoring their robustness in capturing crucial behavioral cues indicative of unsafe driving conditions.

To enable a comprehensive and interpretable assessment of driver safety, we introduce a unified safety score encapsulated as a structured triplet. This triplet combines synchronized outputs of the attention, drowsiness, and emotion models within common temporal windows. Unlike traditional single-value safety scores, this multidimensional representation offers richer insight by simultaneously reflecting attentiveness, fatigue, and emotional affect, thus providing nuanced contextual understanding of driving risk factors.

While these results are promising, it is important to note that we have not yet conducted systematic ablation studies to isolate the impact of each model or to experiment with alternative fusion strategies. The current evaluation focuses on the performance of the individual models and the combined triplet framework. Future work will focus on detailed ablation experiments and fusion optimization to quantify each component’s contribution more precisely.

This section details the experimental evaluation conducted across the three behavioral dimensions, such as attention, facial emotion recognition, and drowsiness detection. All models were trained and validated on publicly available benchmark datasets, ensuring reproducibility. Model performance was assessed using metrics including accuracy, precision, recall, F1-score, and AUC. Where feasible, results were benchmarked against recent state-of-the-art methods. The following subsections provide a detailed account of dataset usage, experimental setups, and quantitative outcomes.

4.1 Driver Drowsiness Detection Accuracy and Limitations

The driver drowsiness detection module was initially trained on the YAWDD and subsequently fine-tuned and evaluated on the NTHU Drowsiness Detection dataset. This approach leverages broad yawning cues from YAWDD while adapting to the multi-modal fatigue indicators present in NTHU.

Our MobileNetV2-LSTM fusion model achieved 93% accuracy on the YAWDD dataset, showcasing robust performance compared to recent baselines: the face-feature vector with sliding

window method at 94.32% [28], a deep feature-based SVM model at 91.7% [26], and a hybrid CNN-LSTM at 92.5% [29] (see Table 1). Fine-tuning on NTHU further improved resilience under real-world variability with accuracy, F1-score, precision, and recall values summarized in Table 2.

To provide deeper insight, the confusion matrix for evaluation on the NTHU dataset is summarized as follows: out of the total samples, 298 instances of alert drivers were correctly identified, 11 were misclassified as drowsy, 118 true drowsy cases were detected accurately, and 21 drowsy cases were missed (classified as alert). This translates to a high true positive rate for both classes, with some moderate bias towards false negatives in detecting subtle drowsiness, a challenging edge case consistent with recent literature.

The use of transfer learning enables the model to combine physiological signals (eye aspect ratio) with spatiotemporal features learned by the deep network, yielding reliable driver state recognition adaptable across different datasets. The fusion of model predictions and EAR-based cues in a voting framework further enhances robustness to variations in camera angle, lighting, and the subtlety of fatigue expressions. These results demonstrate the proposed framework’s effectiveness as a practical and accurate driver drowsiness monitoring solution, achieving performance competitive with state-of-the-art benchmarks and offering a foundation for further development through more sophisticated fusion strategies and ablation studies.

Table 1. Comparison of Driver Drowsiness Detection Accuracy on YAWDD Dataset

Model	Accuracy
Proposed Model (This Paper)	93%
Face-Feature Vector + Sliding Window [28]	94.32%
Deep Learned Features with SVM [26]	91.7%
Hybrid CNN + LSTM [29]	92.5%

Table 2. Evaluation Metrics for Driver Drowsiness Detection

Metric	Value
Accuracy	0.931
F1 Score (Alert)	0.95
F1 Score (Drowsy)	0.88
Recall (EAR method)	0.895
Precision (Drowsy)	0.91
Recall (Drowsy)	0.85

4.2 Facial Emotion Recognition Accuracy and Limitations

Facial expression recognition was performed on the FER2013 dataset. Table 3 shows the accuracy of our ResNet18 model with temporal voting compared to recent deep learning benchmarks.

Table 3. Comparison of Facial Expression Recognition Accuracy on FER2013

Model	Accuracy
Proposed ResNet18 Model (with temporal voting)	93.1%
VGGNet [30]	73.3%
MobileNetV2 [31]	61.0%

Detailed evaluation metrics are listed in Table 4; the ResNet18 model demonstrates a strong precision of 0.9954, recall of 0.9311, and an F1-score of 0.9620, indicating high reliability and balance between false positives and false negatives.

The confusion matrix reveals class-wise prediction distribution: notable sample counts include 84 correctly predicted ‘angry’ emotions with 16 misclassified, 65 ‘disgust’, 95 ‘fear’, 168 ‘happy’, 459 ‘neutral’, 14 ‘sad’, and 42 ‘surprise’ correctly classified. Misclassifications mostly occur between closely related or underrepresented classes such as ‘sad’ and ‘neutral’, reflecting challenges due to class imbalance and subtle expression variations.

The model’s performance underscores the utility of temporal voting combined with transfer learning to enhance robustness against variability in facial expressions typical of real-world driving scenarios. However, limitations include sensitivity to occlusions (e.g., face masks, glasses), extreme head poses, and varied illumination conditions commonly encountered inside vehicles. Additionally, the domain gap between internet-sourced FER2013 images and actual in-vehicle face data may

Table 4. Evaluation Metrics for Facial Emotion Detection Model

Metric	Value
Accuracy	0.9311
Precision	0.9954
Recall	0.9311
F1 Score	0.9620
AUC-ROC	N/A

reduce generalization. Future work should explore domain adaptation and incorporate additional real-world data to address these challenges.

4.3 Driver Attention Model Accuracy and Limitations

The driver attention model employs an LSTM network with an attention mechanism to analyze sequential gaze and head pose features on the NTHU dataset. Our MobileNetV2+LSTM system achieves an accuracy of 96.95%, outperforming previous hybrid CNN-based and LSTM-based approaches, and closely matching the performance of transformer architectures as shown in Table 5. Detailed metrics presented in Table 6 include a precision of 0.9541, recall of 0.9744, F1-score of 0.9642, and an AUC-ROC of 0.9967, underscoring robust generalization to real-world driver monitoring scenarios.

Further analysis of the confusion matrix indicates that the model correctly classified 5,448 true positives and 7,445 true negatives, with 262 false positives and 143 false negatives. This performance reflects strong true positive and true negative rates in identifying both attentive and inattentive driving states, though some misclassifications occur in challenging cases with subtle or ambiguous head pose and gaze cues. While the model demonstrates cutting-edge performance on the NTHU dataset, it is more vulnerable to errors in sequences exhibiting rapid head movements, extreme gaze angles, or partial facial occlusion. Additionally, as with most deep learning systems, precision and recall may decline in substantially different real-world environments or with drivers exhibiting behavioral patterns not well represented in training data. This underscores the need for further research into domain adaptation and in-cabin data augmentation to enhance deployment robustness.

Table 5. Comparison of Driver Attention Detection Accuracy on NTHU Dataset

Model	Accuracy
Proposed Model (MobileNetV2+LSTM, This Paper)	96.95%
Hybrid CNN [32]	95.0%
Transformer-based Model [33]	97.1%
LSTM-based Model [34]	92.2%

Table ?? provides additional metrics for our proposed attention model.

Table 6. Evaluation Metrics for Driver Attention Model

Metric	Value
Accuracy	0.9695
Precision	0.9541
Recall	0.9744
F1 Score	0.9642
AUC-ROC	0.9967

The fusion-based triplet system, which integrates the attention, emotion, and drowsiness modules, achieves a joint window precision of approximately 90.8% across all three tasks. Notably, representing the unified driver state as a three-dimensional triplet rather than reducing it to a single scalar, enhances the interpretability of system output, enabling safety analysts and human operators to identify which specific behavioral dimension (attention, emotion, or drowsiness) most directly contributes to the overall risk assessment. This joint metric represents a more stringent criterion than the individual model accuracies, reflecting the robustness and reliability of the unified monitoring framework in capturing a comprehensive driver state.

It is important to note that this observed performance, while promising, may not fully generalize to in-vehicle deployments due to differences in dataset domains, including variations in

lighting conditions, driver behavior, and sensor positioning. Recent work demonstrates that even high-performing driver monitoring systems can experience a notable drop in accuracy when models are transferred from controlled laboratory or simulator environments to naturalistic in-vehicle settings, owing to factors such as unpredictable lighting, camera placement, and driver variability. For instance, Aravinth et al. (2025) report up to a 10% decrease in multi-modal driver state prediction accuracy after deployment in real-world environments [35].

Future work will focus on conducting extensive ablation studies to isolate and quantify the contributions of each module and fusion strategy, as well as cross-dataset validations to evaluate generalizability. Additionally, the public release of code and pretrained models will facilitate reproducibility, and thorough real-world testing will be conducted under diverse driving conditions. Hardware and software configurations, along with reproducibility protocols, are detailed in the Methods section. Table 7 compares our proposed system with recent state-of-the-art unified driver behavior prediction models, highlighting its competitive performance in a rapidly advancing field.

Table 7. Comparison of Unified Driver Behavior Prediction Models	
Method	Reported Accuracy
Proposed Model (Attention, Emotion, Drowsiness fusion)	90% (triplet fusion)
CNN-LSTM multi-stream fusion [36]	89.5%
Deep CNN + physiological signals [9]	88.9%
Multi-task transformer model [27]	92.0%

4.4 Summary

The unified driver state monitoring system presented herein matches or surpasses published accuracy baselines across its target tasks and datasets. It delivers a reproducible, real-time safety solution built on deep learning architectures combined through a principled fusion strategy. Experimental evaluations demonstrate this system’s efficacy, balanced by transparent acknowledgment of its current limitations and areas for improvement. This work lays a solid foundation for future extensions that will push the field toward safer, more adaptive driver assistance technologies.

Table 8. Performance Comparison of Driver Monitoring Tasks			
Task	Model Used	Accuracy	Voting Applied
Attentiveness Detection	LSTM-RNN	96.9	No
Facial Expression	ResNet-18	93.11	Yes (30-frame)
Drowsiness Detection	CNN + EAR Rule	93.1	Yes (3x10)

5 Conclusion

This study presents a scalable, real-time driver monitoring system that integrates key behavioral assessments, including driver drowsiness detection, attentiveness evaluation, and emotion recognition. Leveraging advanced deep learning architectures such as MobileNetV2, LSTM, and ResNet18, the system captures complex spatial and temporal features from in-cabin video streams. By combining data-driven features with physiological metrics like eye aspect ratio, it achieves reliable detection even under challenging conditions, including varying illumination and occlusions.

Extensive evaluations on benchmark datasets demonstrate high accuracy and strong generalization across diverse scenarios (see Table 8). The unified system uniquely represents driver state as a triplet comprising attention, emotion, and drowsiness scores, which enhances interpretability by enabling analysts to discern the primary contributors to overall driver risk. This modular design offers flexibility for updating individual components and incorporating additional sensors, facilitating adaptation to future needs and environments. Real-time processing capabilities with low inference latency confirm the suitability of this technology for deployment in modern vehicles.

Despite these strengths, the system faces challenges, including performance degradation under severe occlusions and low-light conditions common in real-world applications. Generalization to naturalistic in-vehicle environments remains non-trivial; recent studies report accuracy reductions

of up to 10% due to domain shifts in lighting, driver diversity, and sensor placement. Addressing these limitations through extended multi-modal physiological sensor integration and advanced temporal modeling with transformer architectures is the focus of ongoing work. Overall, the proposed unified driver monitoring framework marks an important advance in proactive driver risk assessment, offering promising potential to enhance road safety through timely and accurate behavioral analysis.

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During the preparation of this work, the author(s) used Grammarly to enhance the readability and language of the manuscript. The tool was used solely for grammar and language suggestions. All content was carefully reviewed and verified by the authors.

Author contributions

Kanika Dhawan: Conceptualization, Methodology, Data Curation, Software, Formal Analysis, Visualization, Writing – Original Draft. Prince Kushwaha: Data Curation, Software, Validation, Investigation, Writing – Review & Editing. Sukhjit Singh Sehra: Supervision, Conceptualization, Project Administration, Resources, Funding Acquisition, Writing – Review & Editing. Jaiteg Singh: Methodology, Validation, Writing – Review & Editing. Sumeet Kaur Sehra: Visualization, Resources, Writing – Review & Editing.

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